CptS 487: Project "Secret Feature"

Compile and Conquer

For our secret feature implementation, we will be focusing on attack B for the final boss fight. In this attack, the boss spawns a ring of slow-moving projectiles around it, which then slowly move out, creating a circle that continues to grow made of the projectiles. Once the projectiles are away from the boss, it shoots out five larger circular projectiles that move faster than the first round and pass through them. As they pass through, they collect the first round of projectiles in front of them as they come in contact with them and bring them along in front. They then reach the end of the screen, where they explode into a grouping of projectiles that seem to have random direction and velocity. Our goal will be to implement each of those three projectile stages as closely as possible.

Our existing game design is well-suited for this task. Two key components we will be discussing that are relevant to this secret feature are collision detection and bullet movement patterns. For those two systems, we used the observer pattern for our entity collisions and the command pattern for entity movement. Since our entity collisions are already set up to notify entities of a collision with any collidable entity, we can use an OnCollide function in our new type of enemy bullets to detect collisions between the bullets. We can also implement the new circular bullets by creating a specialized movement pattern for them that lets us specify the angle.

Luckily, due to those existing game designs, we can easily implement this feature. The only major refactoring we will need to do is with our base enemy bullets, as we need their position to be able to be controlled by the larger bullets that will come after them. We can do this by giving them a ‘controlled’ tag, which will then allow our larger bullets to control them. We give our larger bullets a list of enemy bullets that we will add to when we collide with them, and then iterate through the list and update their positions to be in front of our larger bullet. We will only activate this controlled tag when the smaller bullets collide with the larger one, so no other entity or object will be able to change their position.

The last step is to spawn the flurry of bullets when the larger bullet exits the screen. This can be done by using our existing implementation of an isAlive tag. Currently, that is set to false when any entity leaves the screen bounds. When this tag is set to false on our larger bullets,

meaning they need to be destroyed, we will spawn a flurry of bullets heading in random directions using a movement pattern similar to our circular pattern. Using all of the proposed changes and existing systems and design patterns will allow us to implement this secret feature. Minor refactoring will be needed as described but it will allow for even more flexibility in our codebase in the future.